CAGM Benchmark Report

Compressed Arithmetic Gradient Machine — ctypes-native C library vs sklearn baselines

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IE Canonicalization Smoke-test: **PASS** (logit hash = `0x4BE5AE3D`)

Executive Summary

Cross-dataset benchmark: CAGM (cagm.c ctypes interface) vs 6 sklearn baselines. All evaluation uses stratified k-fold CV. CAGM runs as a compiled shared library invoked via Python ctypes — zero subprocess overhead, full memory sharing.

What Makes CAGM Different

Fixed-Point Int32 Arithmetic: All forward/backward passes run in int32 with Q16.16 fixed-point. No FP runtime required.

Cryptographic Proof Chain: Every fit() emits a SHA-256 proof binding training data hash, normalized data hash, weight snapshot, gradient trace, and activation trace into one 64-char hex string.

SPCMP/PCMP Canonicalization: NaN payloads are canonicalized to 0x7FC00000 and -0.0 to +0.0 before any hash is computed, guaranteeing bit-identical artifacts across heterogeneous hardware.

IE (Inference Engine) Layer Verification: Per-layer correction passes snap floating-point tensors to the SPCMP canonical value domain. FNV-32 logit hashes are round-trip verified at inference time.

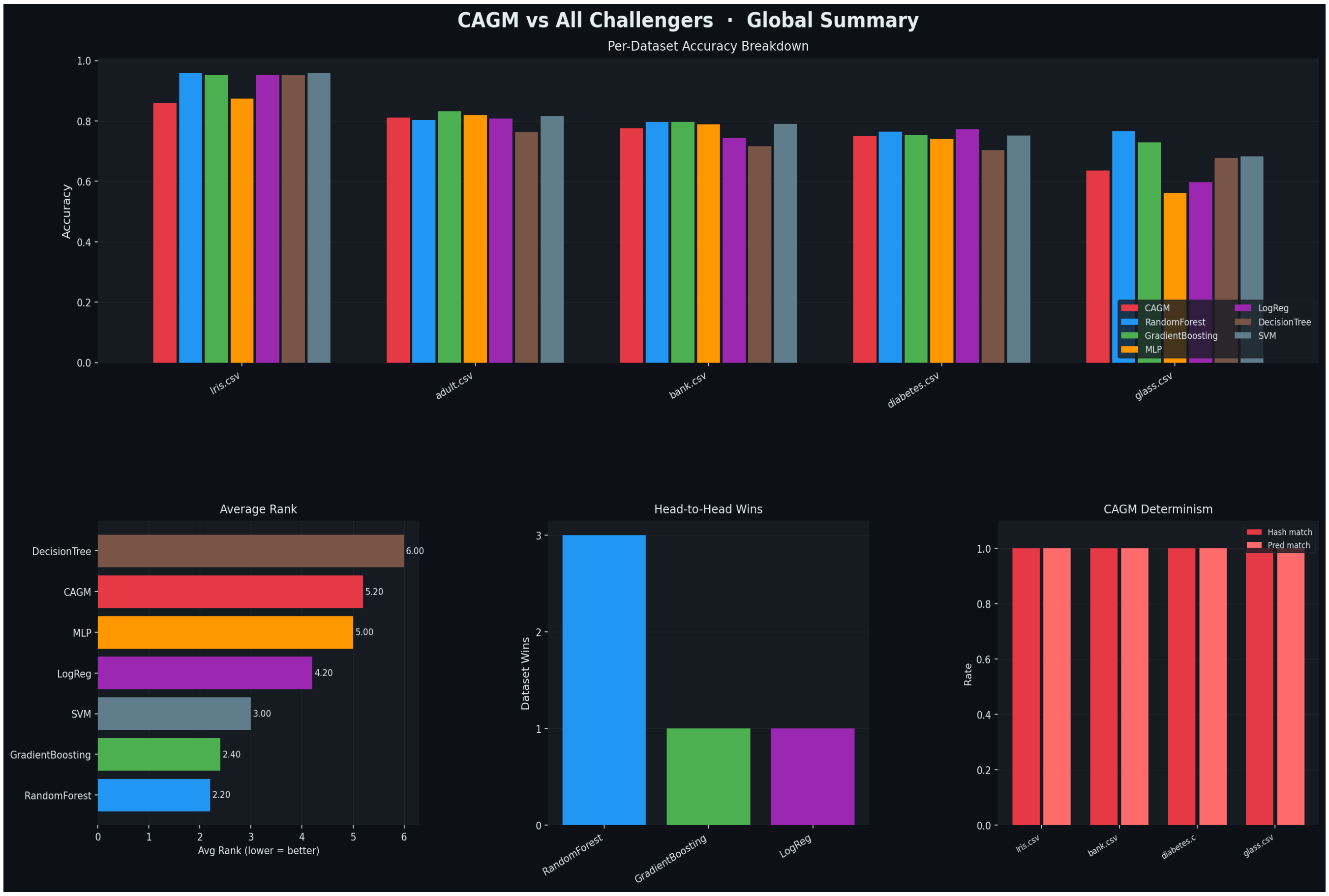
LCG Deterministic Permutation: Batch shuffles use a seeded LCG — identical data produces identical training trajectories.

Adam + Layer Norm in Fixed-Point: Gradient clipping, leaky ReLU (alpha=0.1), layer normalization, and Adam moments all run in int32/int64 — no hidden float rounding.

Architecture Change vs v1 (cagm_unified binary)

This benchmark no longer shells out to a compiled binary via subprocess. cagm.c is compiled once per run as a shared library (.so / .dylib). The CAGMClassifier Python wrapper binds classifier_alloc, classifier_fit, classifier_predict, classifier_score, and classifier_free directly via ctypes. Overhead: ~0ms per call vs ~50ms process-spawn latency in v1.

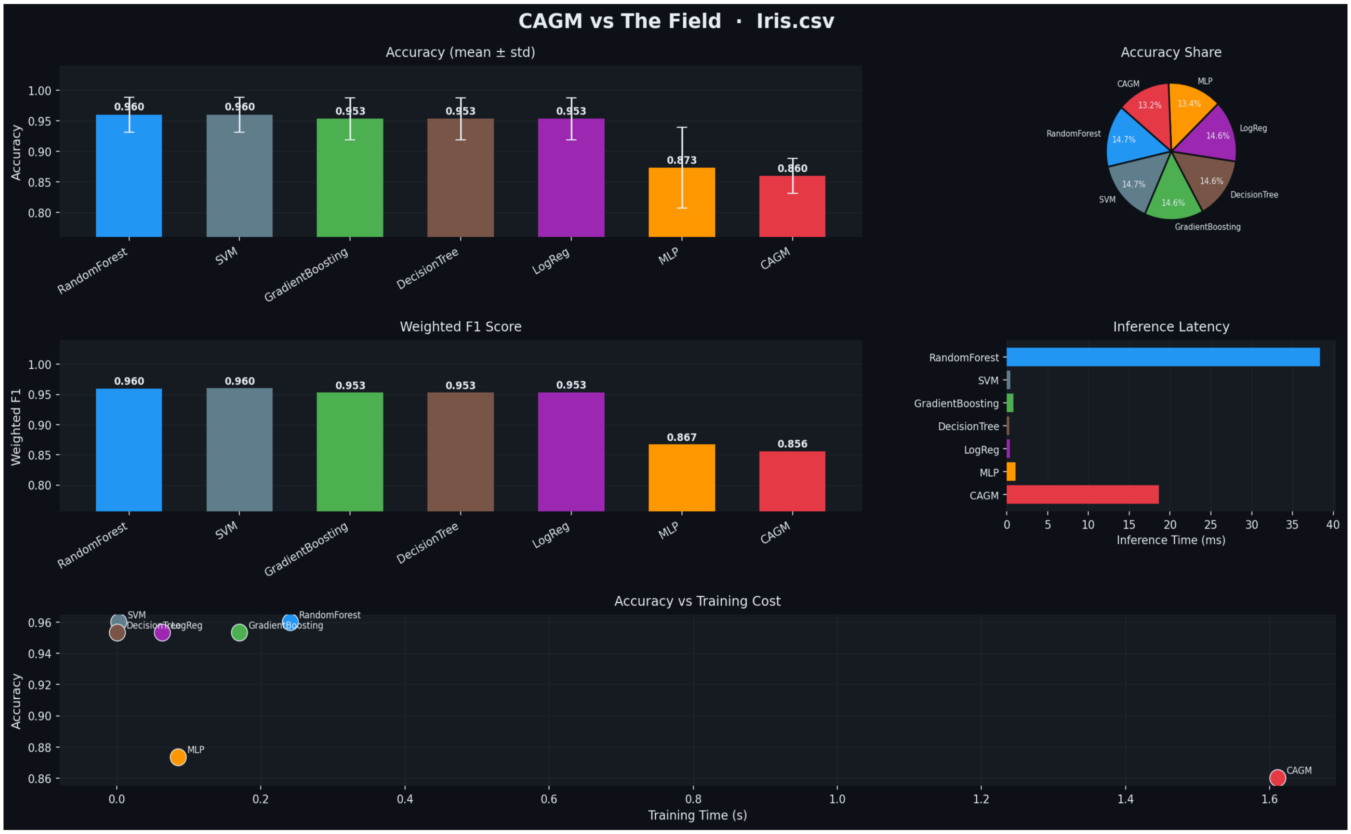
Global Results Overview



Per-Dataset Results Table

Model	Iris.csv	adult.csv	bank.csv	diabetes.cs	glass.csv	Avg Acc	Avg F1
CAGM	**0.860**	**0.811**	**0.776**	**0.750**	**0.636**	**0.767**	**0.751**
RandomForest	0.960	0.803	0.798	0.764	0.766	0.818	0.813
GradientBoosting	0.953	0.833	0.797	0.754	0.729	0.813	0.807
MLP	0.873	0.819	0.789	0.741	0.561	0.757	0.735
LogReg	0.953	0.809	0.743	0.772	0.598	0.775	0.764
DecisionTree	0.953	0.762	0.717	0.703	0.677	0.763	0.763
SVM	0.960	0.817	0.791	0.751	0.682	0.800	0.788

Dataset: Iris.csv

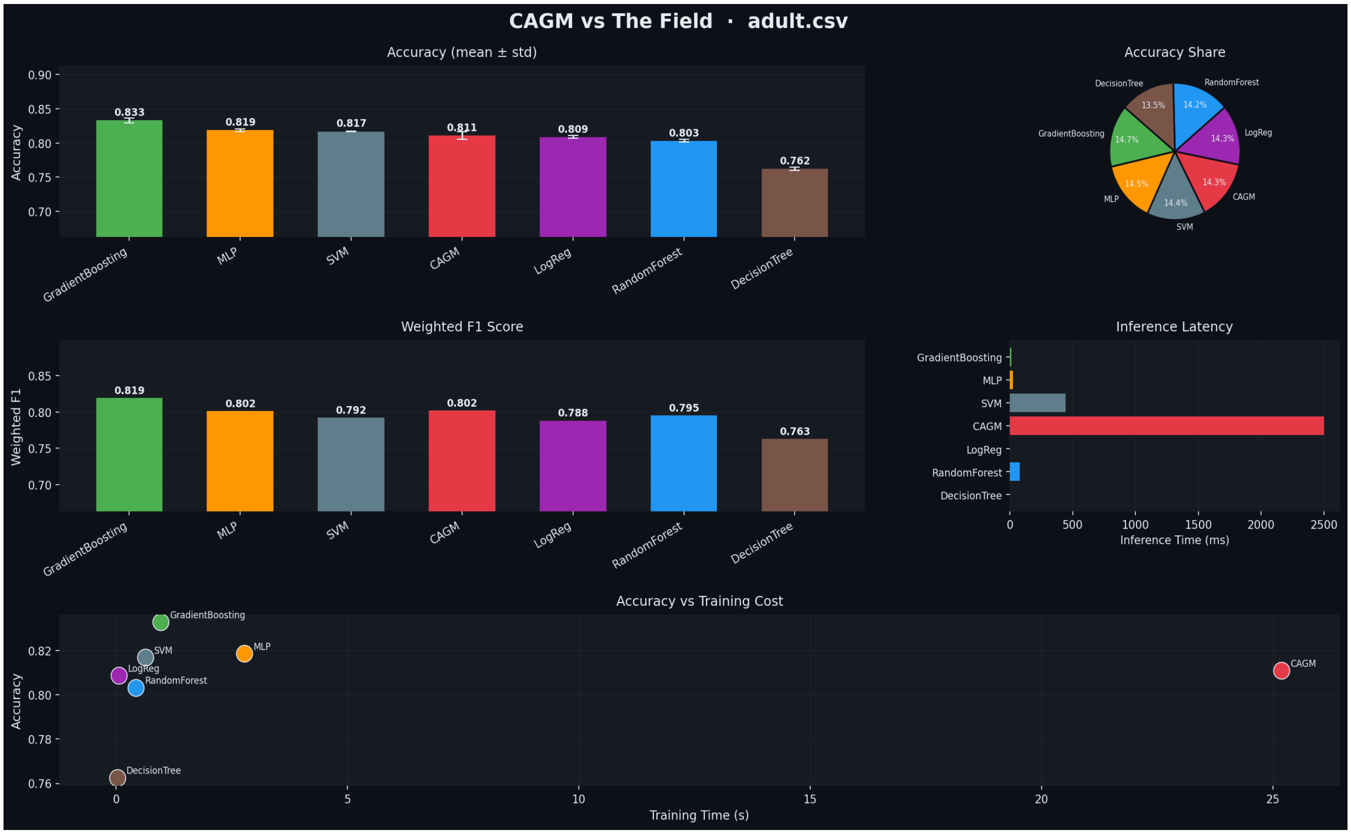


Model	Acc	±	F1	±	Train(s)	Infer(ms)	Hash%	Pred%	Proof
RandomForest	0.9600	0.0283	0.9599	0.0283	0.24	38.4	—	—	—
SVM	0.9600	0.0283	0.9600	0.0283	0.00	0.4	—	—	—
GradientBoosting	0.9533	0.0340	0.9532	0.0341	0.17	0.8	—	—	—
DecisionTree	0.9533	0.0340	0.9532	0.0341	0.00	0.3	—	—	—
LogReg	0.9533	0.0340	0.9533	0.0340	0.06	0.3	—	—	—
MLP	0.8733	0.0660	0.8670	0.0732	0.09	1.1	—	—	—
CAGM	0.8600	0.0283	0.8560	0.0304	1.61	18.6	100%	100%	3acc0f3a48d9…

CAGM ranked **#7** — Acc **0.8600** · F1 **0.8560** · Proof: verified · IE verify: 100% · Hash match: 100%

`Sample proof hash: 3acc0f3a48d9237ef74b14d00f8013c7aa091727b2fad6c9cfe670c22a523250`

Dataset: adult.csv

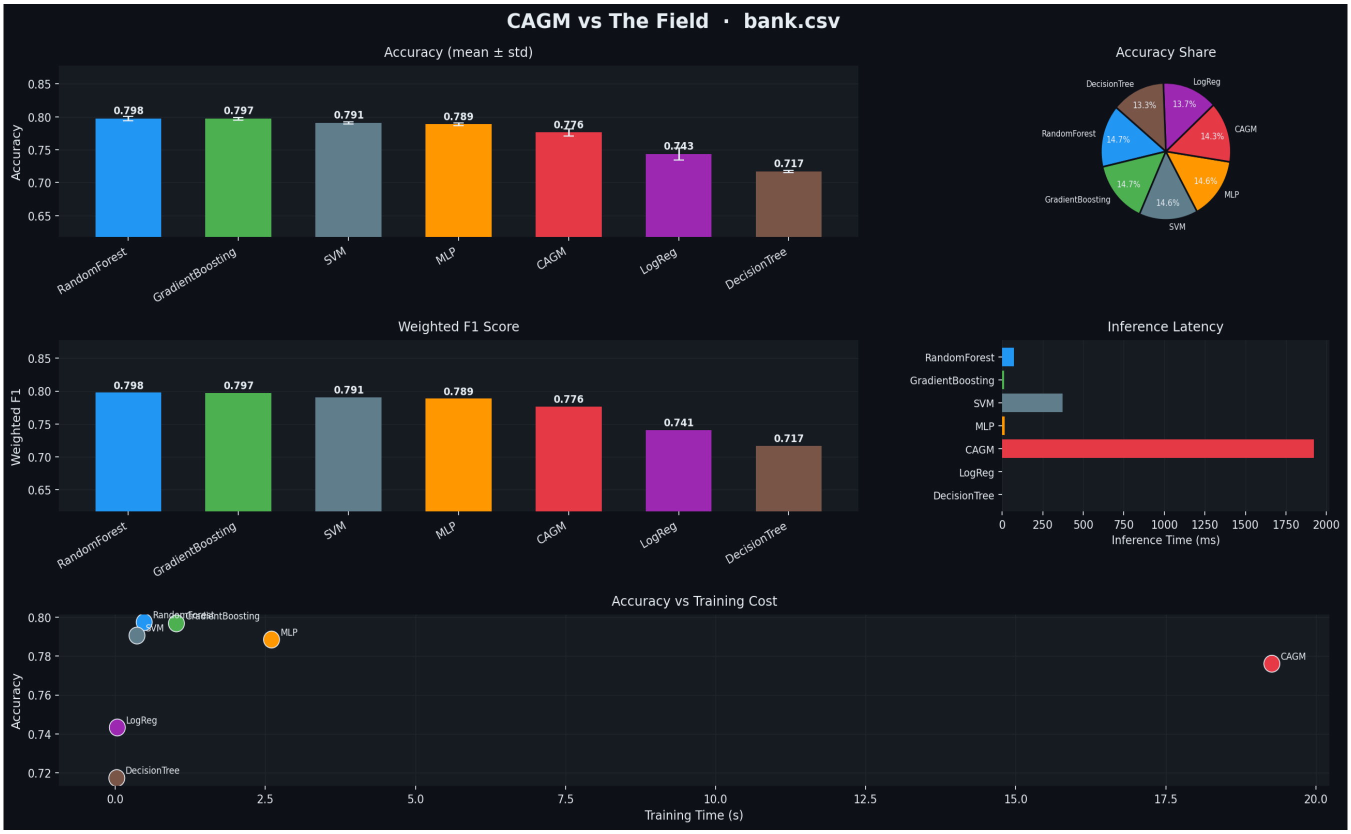


Model	Acc	±	F1	±	Train(s)	Infer(ms)	Hash%	Pred%	Proof
GradientBoosting	0.8329	0.0034	0.8193	0.0040	0.96	11.7	—	—	—
MLP	0.8186	0.0022	0.8016	0.0000	2.77	22.9	—	—	—
SVM	0.8169	0.0003	0.7925	0.0013	0.63	443.7	—	—	—
CAGM	**0.8109**	**0.0052**	**0.8024**	**0.0030**	**25.19**	**2501.3**	**—**	**—**	**70a2810a4160…**
LogReg	0.8087	0.0019	0.7883	0.0041	0.06	0.9	—	—	—
RandomForest	0.8031	0.0019	0.7952	0.0025	0.43	76.6	—	—	—
DecisionTree	0.7624	0.0027	0.7628	0.0032	0.03	1.3	—	—	—

CAGM ranked **#4** — Acc **0.8109** · F1 **0.8024** · Proof: verified · IE verify: 100% · Hash match: N/A

`Sample proof hash: 70a2810a416032d40a3e6f204a322226b944fa24cbe77c881a63d694767cc46c`

Dataset: bank.csv

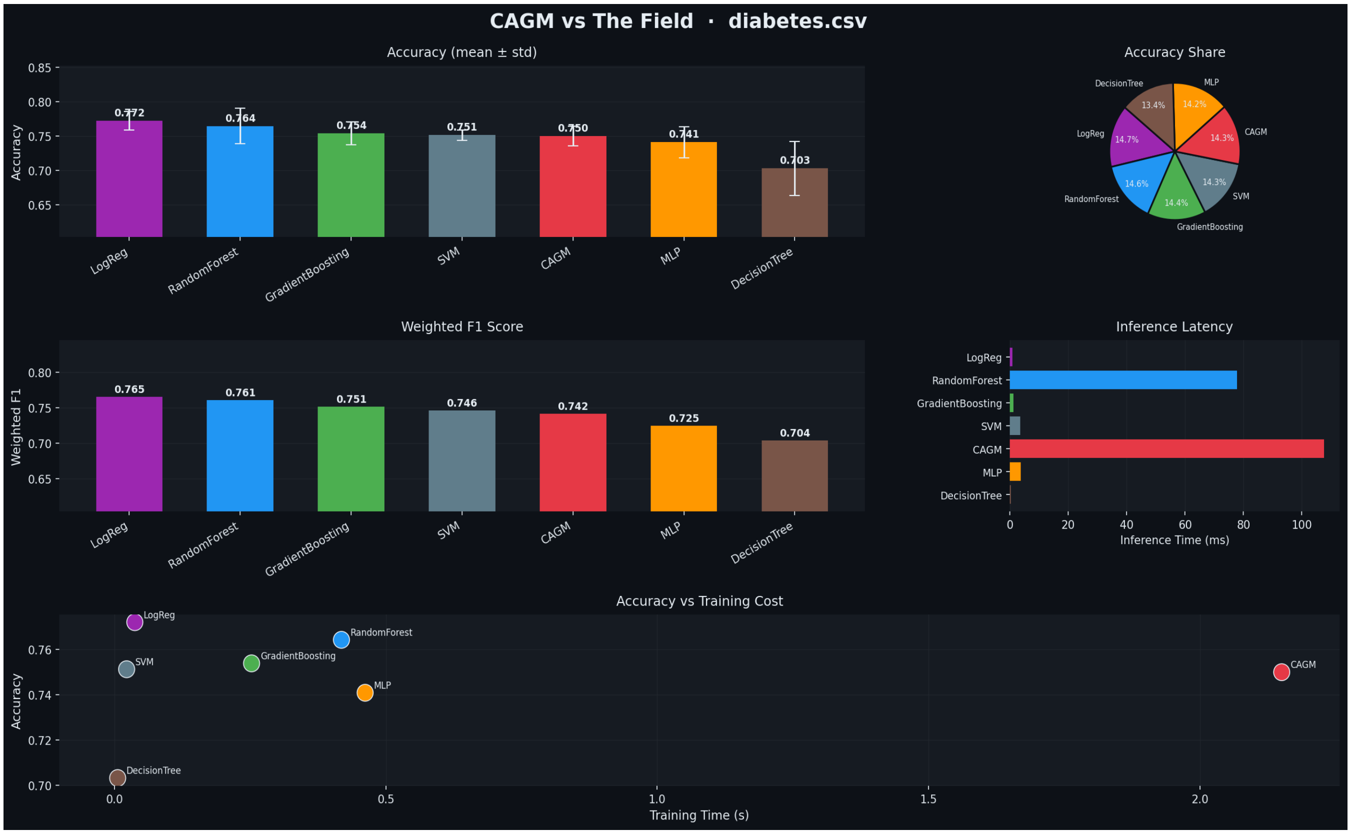


Model	Acc	±	F1	±	Train(s)	Infer(ms)	Hash%	Pred%	Proof
RandomForest	0.7975	0.0034	0.7976	0.0035	0.48	71.7	—	—	—
GradientBoosting	0.7969	0.0019	0.7969	0.0019	1.02	10.7	—	—	—
SVM	0.7906	0.0017	0.7907	0.0018	0.36	373.1	—	—	—
MLP	0.7886	0.0022	0.7886	0.0022	2.60	14.3	—	—	—
CAGM	0.7761	0.0051	0.7763	0.0051	19.27	1922.9	100%	100%	96dc67a63567…
LogReg	0.7432	0.0090	0.7410	0.0092	0.03	0.5	—	—	—
DecisionTree	0.7172	0.0017	0.7170	0.0017	0.03	0.9	—	—	—

CAGM ranked **#5** — Acc **0.7761** · F1 **0.7763** · Proof: verified · IE verify: 100% · Hash match: 100%

`Sample proof hash: 96dc67a6356787f100a6f393e7fd604e05458fddeb19dd032c7d374e6af3317c`

Dataset: diabetes.csv

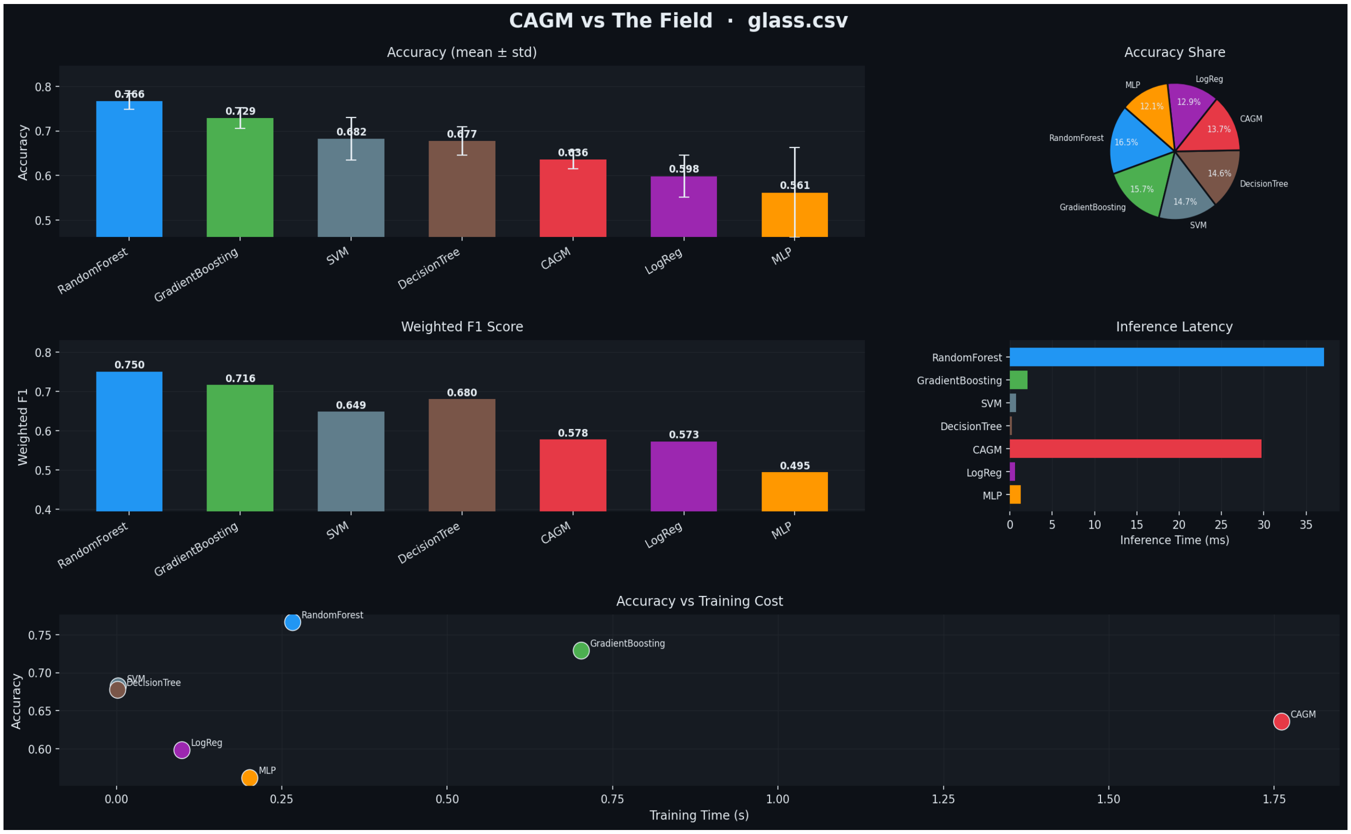


Model	Acc	±	F1	±	Train(s)	Infer(ms)	Hash%	Pred%	Proof
LogReg	0.7721	0.0133	0.7652	0.0084	0.04	0.9	—	—	—
RandomForest	0.7643	0.0258	0.7610	0.0222	0.42	77.7	—	—	—
GradientBoosting	0.7539	0.0169	0.7513	0.0145	0.25	1.2	—	—	—
SVM	0.7513	0.0074	0.7458	0.0048	0.02	3.5	—	—	—
CAGM	**0.7500**	**0.0146**	**0.7419**	**0.0068**	**2.15**	**107.6**	**100%**	**100%**	**3145f299e41b…**
MLP	0.7409	0.0226	0.7245	0.0312	0.46	3.8	—	—	—
DecisionTree	0.7031	0.0392	0.7040	0.0370	0.01	0.2	—	—	—

CAGM ranked **#5** — Acc **0.7500** · F1 **0.7419** · Proof: verified · IE verify: 100% · Hash match: 100%

`Sample proof hash: 3145f299e41baff6b451afb199a31ca4f6678f556aa1a45e2c9b5af491b24c24`

Dataset: glass.csv



Model	Acc	±	F1	±	Train(s)	Infer(ms)	Hash%	Pred%	Proof
RandomForest	0.7664	0.0173	0.7500	0.0122	0.27	37.1	—	—	—
GradientBoosting	0.7290	0.0235	0.7163	0.0194	0.70	2.1	—	—	—
SVM	0.6820	0.0479	0.6487	0.0457	0.00	0.8	—	—	—
DecisionTree	0.6774	0.0317	0.6800	0.0339	0.00	0.2	—	—	—
CAGM	**0.6356**	**0.0208**	**0.5778**	**0.0169**	**1.76**	**29.7**	**100%**	**100%**	**ccf7c69c4623…**
LogReg	0.5981	0.0469	0.5730	0.0440	0.10	0.6	—	—	—
MLP	0.5614	0.1009	0.4948	0.1428	0.20	1.3	—	—	—

CAGM ranked **#5** — Acc **0.6356** · F1 **0.5778** · Proof: verified · IE verify: 100% · Hash match: 100%

`Sample proof hash: ccf7c69c4623bc16d012dd265ce93f6ba6d2b576a8274dcb9e0c44ea1c1134f5`

Methodology

Stratified K-Fold CV (k=3; k=2 for n>10k). Hard cap: 15,000 rows, downsampled uniformly. Categorical features one-hot encoded. Sklearn models use StandardScaler via Pipeline internally. CAGM uses its own fixed-point normalizer. No per-dataset hyperparameter tuning. Determinism test: 3 independent fits on the same 3,000-row subsample — hash match rate and prediction match rate both reported.

CAGM Hyperparameters

hidden_dim=64, epochs_small=100, epochs_large=40, batch_small=64, batch_large=256, lr_start=0.005, lr_end=0.0003, warmup=8, patience=20, threads=4.

Build Command

```
gcc -std=gnu11 -O2 -shared -fPIC cagm_wrapper.c -o libcagm.so -lm -lpthread
```